MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 4 09/09/2021

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We will discuss several questions from Chapter 4 in Ross' book.

Tutorial Questions

(1) Let X be a binomial random variable with parameters n and p. Show that

$$\mathbb{E}\left(\frac{1}{X+1}\right) = \frac{1-(1-p)^{n+1}}{(n+1)p}.$$

(2) Each of 500 soldiers in an army company independently has a certain disease with probability $1/10^3$. This disease will show up in a blood test, and to facilitate matters, blood samples from all 500 soldiers are pooled and tested.

(a) What is the (approximate) probability that the blood test will be positive (that is, at least one person has the disease)?

Suppose now that the blood test yields a positive result.

(b) What is the probability, under this circumstance, that more than one person has the disease?

One of the 500 people is Jones, who knows that he has the disease.

(c) What does Jones think is the probability that more than one person has the disease?

Because the pooled test was positive, the authorities have decided to test each individual separately. The first $i-1$ of these tests were negative, and the ith one—which was on Jones—was positive.

(d) Given the preceding scenario, what is the probability, as a function of i, that any of the remaining people have the disease?

(3) Suppose that 10 balls are put into 5 boxes, with each ball independently being put in box i with probability p_i, $\sum_{i=1}^{5} p_i = 1$.

(a) Find the expected number of boxes that do not have any balls.

(b) Find the expected number of boxes that have exactly 1 ball.

(4) There are k types of coupons. Independently of the types of previously collected coupons, each new coupon collected is of type i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. If n coupons are collected, find the expected number of distinct types that appear in this set. (That is, find the expected number of types of coupons that appear at least once in the set of n coupons.)

Sanity checks: What is your answer when $n = 1$? When $n = 2$? When $n \to \infty$? Why do they make sense?

(5) If X is a geometric random variable, show analytically that

$$P\{X = n+k | X > n\} = P\{X = k\}$$

(6) Find the expected value of the sum obtained when n fair dice are rolled.

(7) An urn initially contains one red and one blue ball. At each stage, a ball is randomly chosen and then replaced along with another of the same colour. Let X denote the selection number of the first chosen ball that is blue. For instance, if the first selection is red and the second blue, then X is equal to 2.

(a) Find $P\{X > i\}$, $i \geq 1$.

(b) Show that, with probability 1, a blue ball is eventually chosen. (That is, show that $P\{X < \infty\} = 1$.)

(c) Find $\mathbb{E}[X]$.

(8) A group of 6 female managers and 19 male managers apply for an assignment. A random sample of 5 people is drawn from a hat (without replacement). What is the probability 4 of the chosen managers will be female and 1 will be male and the cumulative distribution?

Assignment Questions

(1) An interviewer is given a list of people she can interview. If the interviewer needs to interview 5 people, and if each person (independently) agrees to be interviewed with probability 23 , what is the probability that her list of people will enable her to obtain her necessary number of interviews if the list consists of
(a) 5 people and
(b) 8 people?
For part (b), what is the probability that the interviewer will speak to exactly
(c) 6 people and
(d) 7 people on the list?

(2) An urn contains N white and M black balls. Balls are randomly selected, one at a time, until a black one is obtained. If we assume that each ball selected is replaced before the next one is drawn, what is the probability that
(a) exactly n draws are needed?
(b) at least k draws are needed?

(3) Suppose that independent trials, each having probability p, $0 < p < 1$, of being a success are performed until a total of r successes is accumulated. If we let X equal the number of trials required, then

$$P\{X = n\} = \binom{n-1}{r-1} p^r (1-p)^{n-r},$$

for $n = r, r+1, \cdots$. A random variable X whose probability mass function is given as above is said to be a *negative binomial* random variable with parameters (r, p). Note that a geometric random variable is just a negative binomial with parameter $(1, p)$.
(a) Check that $\sum_{n=r}^{\infty} P(X = n) = 1$.
(b) Prove that $\mathbb{E}(X) = r/p$ and $Var(X) = r(1-p)/p^2$.
(c) If independent trials, each resulting in a success with probability p, are performed, what is the probability of r successes occurring before m failures?
(d) Find the expected value and the variance of the number of times one must throw a die until the outcome 1 has occurred 4 times.

(4) Suppose that a batch of 100 items contains 6 that are defective and 94 that are not defective. If X is the number of defective items in a randomly drawn sample of 10 items from the batch, find
(a) $P\{X = 0\}$ and
(b) $P\{X > 2\}$.

(5) A purchaser of electrical components buys them in lots of size 10. It is his policy to inspect 3 components randomly from a lot and to accept the lot only if all 3 are nondefective. If 30 percent of the lots have 4 defective components and 70 percent have only 1, what proportion of lots does the purchaser reject?

(6) Find the expected total number of successes that result from n trials when trial i is a success with probability p_i, $i = 1, \cdots, n$.

(7) Derive an expression for the variance of the number of successful trials in the question above, and apply it to obtain the variance of a binomial random variable with parameters n and p, and of a hypergeometric random variable equal to the number of white balls

chosen when n balls are randomly chosen from an urn containing N balls of which m are white.